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Numerical Analysis Python Programs
Fixed-Point Iteration (General Iterative Method) 2.0

This document provides:

- **A short intro to Fixed-Point Iteration**
- **The program's problem statement (assignment prompt and requirements)**
- **The code description and implementation details.**

1. Introduction to Fixed-Point Iteration

- **Fixed-Point Iteration** (or General Iterative Method) is a **fundamental numerical technique** for finding **real roots** of a non-linear equation $\mathbf{f(x) = 0}$. By algebraically rearranging f(x) = 0 into the equivalent **fixed-point form**:

$$x = g(x)$$

we construct the **iterative sequence**:

$$x_{k+1} = g(x_k), \quad k = 0, 1, 2, \ldots$$

- If g is **continuous** and the **generated sequence** converges, its limit x^* satisfies $x^* = g(x^*)$, which is both **a fixed point of $g(x)$ and a root of $f(x)$**.
- **Convergence and Stability**:
  - **Contraction Mapping Condition**: By the Banach **Fixed-Point Theorem**, convergence is **guaranteed** in an **interval I** containing x^* if $g(x)$ is a contraction, which holds if $|g'(x)| \leq \lambda < 1$ for all $x \in I$.
  - **Order of Convergence**: The asymptotic error satisfies $\lim_{k\to\infty} \frac{|e_{k+1}|}{|e_k|} = |g'(x^*)|$. When $0 < |g'(x^*)| < 1$, the method exhibits **linear convergence** ($p = 1$) with asymptotic error **constant** $\mathbf{C = |g'(x^*)|}$.

2. Problem Statement

- **Non-linear Equation:**

$$f(x) = \cos(x) - x = 0$$

- **Rearrangement Formulation:**

$$x = g(x), \quad \text{where } g(x) = \cos(x)$$

- **Requirements:**

1. **Fixed-Point Iteration:** We must implement $x_{k+1} = g(x_k)$ in `Python` starting from the **initial value** $x_0 = 0$, with stopping **tolerance** $|x_{k+1} - x_k| < 10^{-6}$ and a **maximum limit of 100 iterations**. We will store all **approximations** x_k into a suitable data structure (list).

2. **Comparative Curve Plot:** Using **`matplotlib`**, we plot the curves $y = g(x) = \cos(x)$ (in `green`) and $y = x$ (in **black**) on the same **coordinate axes,** to show their **intersection point** (**fixed point**). The figure **must include axis labels**, **a title**, **a legend**, and **a grid**.

3. **Rate & Constant Estimation**: Utilizing the formulas:

$$p_k = \frac{\ln(e_{k+1}/e_k)}{\ln(e_k/e_{k-1})} \quad \text{and} \quad C_k = \frac{e_{k+1}}{e_k^{p_k}}$$

we estimate the **numerical order** of **convergence** $\mathbf{p_k}$ and the **asymptotic error constant** $\mathbf{C_k}$.

3. Code Description & Implementation Details

- **Initial Setup & Helper Functions**:
  - We import `numpy` and `matplotlib.pyplot`, setting **`tol = 1e-6`** and **`max_iter = 100.`**
  - Both $f(x) = \cos(x) - x$ and $g(x) = \cos(x)$ are declared using `np.cos(x)`.

- **The Iteration Function (`FPI(x0)`):**
  - **Sequence Storage**: We store initial point x_0 and all **subsequent steps** inside **`x_k = [x0]`**.

- **Step Evolution:** In **each iteration loop** i, the **next point** $x_{next} = g(x_0)$ is computed and **appended to** `x_k`. When the **consecutive difference** $|x_{\text{next}} - x_0|$ **drops below** `tol`, the loop **terminates early** via **break**; otherwise, x_0 is updated to x_{next}.

- **Execution:** Executing `x_k = FPI(0)` computes the **root sequence** starting **strictly at** $x_0 = 0$.

- **Geometric curve visualization**:

  - A **continuous domain** $x \in [-5, 5]$ is sampled with **300 points** via `np.linspace(-5, 5, 300)`.

  - `plt.plot(x, np.cos(x), 'g', label='y = g(x) = cosx')` renders the **cosine function** in green, while `plt.plot(x, x, 'k', label='y = x')` renders the **identity line** in **black**.

  - The **visual intersection** clearly **highlights** the **unique real fixed point** $x^* \approx 0.739085$.

- **Asymptotic Convergence Calculation**:

  - The consecutive **step errors** are returned **directly by the function** via `x_k, errors = FPI(0)`.
  - The **last three consecutive errors** are **extracted using negative indices**:
    $e_{k-1} = \text{errors}[-3]$, $e_k = \text{errors}[-2]$, $e_{k+1} = \text{errors}[-1]$

  - **Substituting these into the logarithmic formulas** yields $p_k \approx 1.0$ and $C_k \approx 0.6736$.

  - **This confirms linear convergence** and **numerically matches the theoretical asymptotic rate**:

$$C = |g'(x^*)| = |-\sin(x^*)| = \sin(0.739085) \approx 0.6736$$